

Stock Forecasting Tool

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— Welcome to the Prophet Forecasting App: A Data Science Exploration —

by Shane Peterson

This tool demonstrates the application of the Prophet forecasting model for predicting short-term price movements in stocks. This project focuses on showcasing the following data science skills:

- **Time Series Forecasting:** Utilizing the Prophet model to predict stock prices.
- **Hyperparameter Tuning:** Optimizing model performance through hyperparameter tuning. (see About section below)
- **Model Evaluation:** Assessing model performance using appropriate metrics (e.g., SMAPE, RMSE). (see Accuracy Metrics section below)

The app includes visualizations such as line charts, moving averages, and Bollinger Bands to provide context for the forecast.

Note on Data: *This app uses a limited historical data source for demonstration purposes. Refer to the 'About' section for details on model validation and data limitations.*

Disclaimer: This app is for educational and demonstrative purposes only. It is not a financial recommendation and should not be used for actual trading decisions.

For more technical details please refer to the About section and the Appendix.

— Choose a Stock —

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Enter Symbol (Ticker List in Appendix)

spy

-- SPDR S&P 500 ETF TRUST Data Loaded. --

	Symbol	Current Price	Daily % Change	YTD % Change	52-Week High	52-Week Low	Current Volume
0	SPY	628.81	0.01%	7.56%	628.81	496.48	59,760

— Model Configuration —

Training Data Duration (Trading Days)

500

30500

Trend Flexibility

5

110

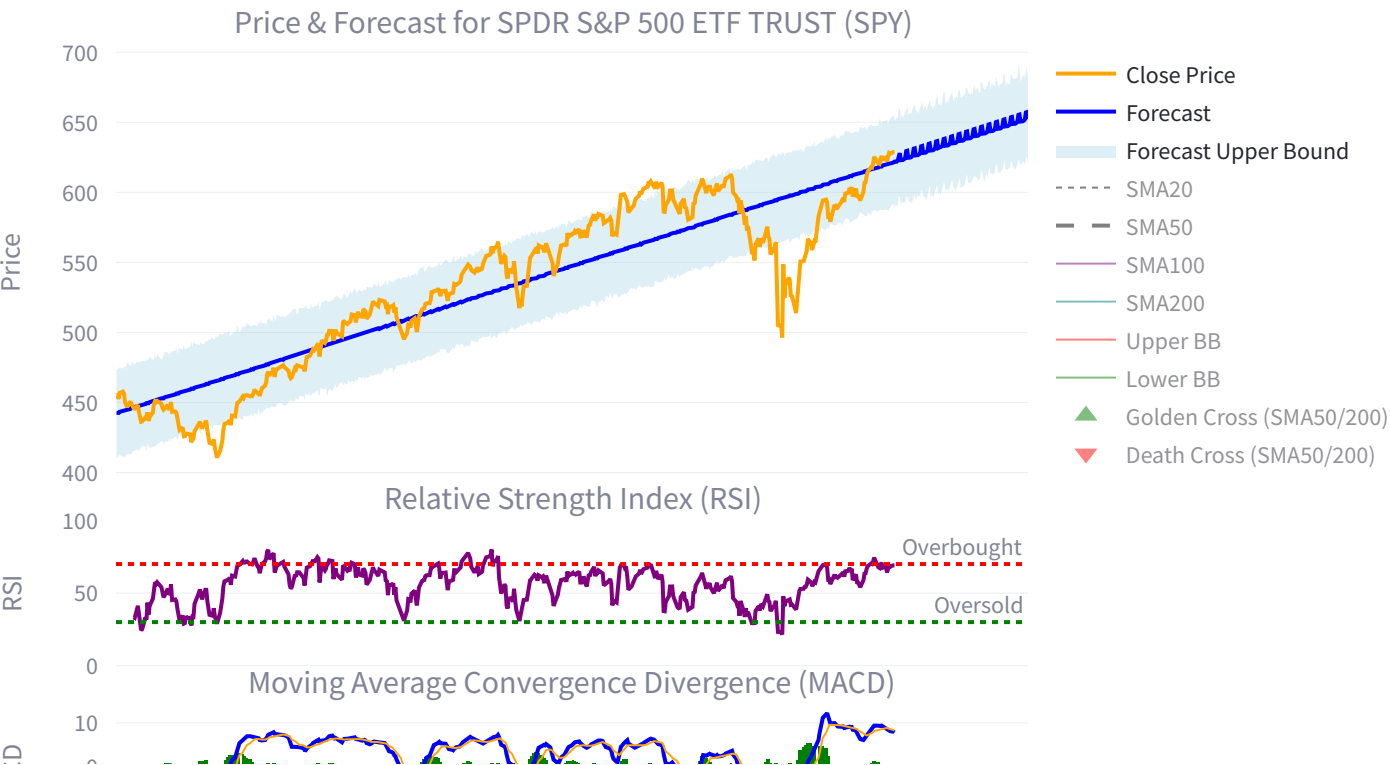
Seasonality Strength

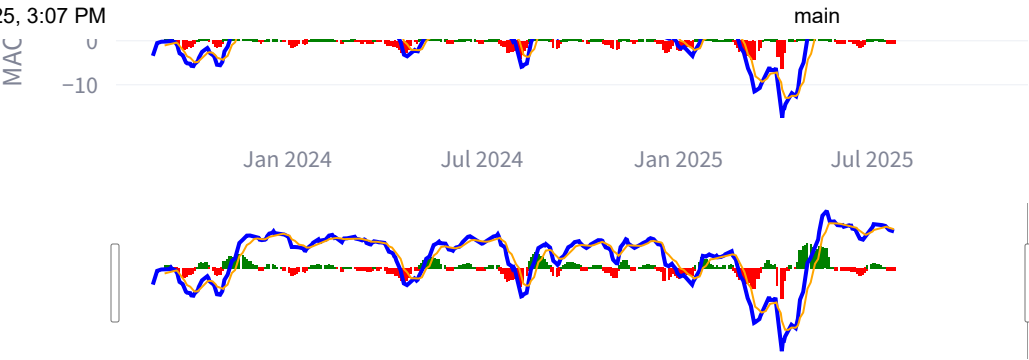
5

110

All models trained successfully! Final model uses winsorized data.

Stock Price Forecast and Indicators for SPDR S&P 500 ETF TRUST (SPY)





10-Trading-Day Forecast Summary for SPY

The forecast predicts the price of SPY will be \$624.39 on 2025-08-01.

This represents a -0.70% decrease from the last known price.

With 80% confidence, the price is expected to be between \$592.28 and \$655.41.

— Predicted Accuracy —

Model Accuracy

97.67%

— Business Intelligence KPIs —

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Price Movement Forecasts

	1-Day Forecast Price Change	1-Day Forecast Price	7-Day Forecast Price Change	7-Day Forecast Price
Value	-1.18% decrease	\$621.41	-0.82% decrease	\$623.66

Volume Trends

	Current/Last Known Volume	Average Daily Volume (Historical)	% Diff from Average Volume
Value	59,763,205	65,509,449	-8.77%

- **Significance:** Volume indicates market interest and liquidity.
- Higher volumes during price movements can confirm the strength of a trend.
- A significant difference between recent and average volume might suggest unusual trading activity.
- **Interpreting Percentage Difference:**
 - **High Positive (e.g., > +20%):** Suggests unusually strong buying or selling interest, often confirming a price trend.
 - **High Negative (e.g., < -20%):** Indicates declining interest or lower liquidity, which can signal weakening trends.
 - *Note: The exact 'significant' threshold can vary greatly depending on the stock's typical liquidity and market conditions.*

Volatility Assessment

Annualized Volatility: 16.53%

Volatility Rank: Low

(Higher volatility indicates greater price fluctuation risk and potential for larger daily swings.)

Market Correlation

Correlation to S&P 500 (SPY): 1.00

- A value close to +1 means strong positive correlation, -1 strong negative correlation, 0 no correlation.
- **Implications:**
 - **Positive (+1):** Stock generally moves *with* the market. Offers limited diversification; exposed to broader market risks.
 - **Negative (-1):** Stock generally moves *opposite* the market. Can offer diversification/hedging benefits (rare for individual stocks).
 - **Near Zero (0):** Stock moves *independently* of the market. Good for diversification; performance tied to company-specific factors.

— Model Iterations and Performance (Narrative & KPIs) —

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The tables below illustrate the methodical and iterative approach to model refinement and performance optimization, a critical skill in data science. We begin with a **Baseline Model** using raw data, then introduce a **Winsorized Model** to address data challenges posed by outliers, demonstrating the implementation of effective data preprocessing solutions. Your script then selects the best-performing model between the Baseline and Winsorized models for **hyperparameter tuning**, resulting in the **Final Model**. Each stage is rigorously evaluated using cross-validation and a comparison of performance metrics (SMAPE, RMSE, MAE, MSE), clearly demonstrating the tangible improvements gained at each step. This process highlights a comprehensive data science workflow, from identifying data-driven problems and implementing solutions to optimizing model performance and validating improvements for enhanced predictive accuracy and business value.

Baseline Model

	mae	mse	rmse	smape
Baseline Model	26.9675	1,162.3326	34.093	0.046

This initial model establishes a performance benchmark using raw historical data. It provides a baseline understanding of forecasting accuracy before applying any advanced data preprocessing or tuning techniques.

Winsorized Model

	mae	mse	rmse	smape
Winsorized Model	23.3109	903.1048	30.0517	0.0399

Impact of Winsorization: This model applies Winsorization to mitigate the impact of outliers. In this instance, by addressing outliers, the Winsorized Model improved predictive accuracy. We observed a **11.85% reduction in RMSE**, a **13.56% reduction in MAE**, and a **13.31% reduction in SMAPE** compared to the Baseline Model. This demonstrates the potential role of data preprocessing in building more robust and reliable forecasts, leading to more confident decision-making by mitigating the impact of extreme price fluctuations.

Final Model (Hyperparameter Tuned)

	mae	mse	rmse	smape
Final Model	13.6414	378.5853	19.4573	0.0233

Impact of Hyperparameter Tuning: The Final Model, optimized through rigorous hyperparameter tuning (based on the superior performance of the **Winsorized Model**), achieved further performance gains. We saw an additional **35.25% reduction in RMSE**, a **41.48% reduction in MAE**, and a **41.70% reduction in SMAPE** compared to the Winsorized Model. This fine-tuning process ensures the model precisely captures underlying patterns, delivering highly accurate forecasts that directly translate into improved decision quality and reduced financial risk.

In the context of time series forecasting, 'error' refers to the difference between the actual value of a variable at a specific point in time and the value predicted by a forecasting model. In this case, the metrics will specifically measure the error between the stock's closing price and the forecast trained on the closing price.

- **MAE (Mean Absolute Error) KPI:** A MAE of **\$13.64** implies that, on average, the model's predictions are off by approximately **\$13.64**. This is a direct measure of prediction accuracy in currency units.
- **SMAPE (Symmetric Mean Absolute Percentage Error) KPI:** A SMAPE of **2.33%** means that, on average, the model's predictions are **2.33%** off from the actual values. This provides a normalized, business-friendly view of percentage accuracy.
- **MSE (Mean Squared Error) KPI:** This squares the errors, giving more weight to larger errors. A lower MSE indicates better accuracy. While less intuitive for direct business interpretation, it's a critical metric for model optimization.
- **RMSE (Root Mean Squared Error) KPI:** The RMSE of **\$19.46** suggests that the model's predictions can deviate from the actual values by up to **\$19.46** in some cases. Being in the same units as the stock price, it offers a tangible measure of typical prediction error.

Beyond statistical measures, these metrics translate directly into business impact. For instance, a lower Mean Absolute Error (MAE) signifies that, on average, our forecasts are closer to the actual stock price. This precision is critical as it directly reduces potential financial risk by providing more accurate price expectations. Similarly, the Root Mean Squared Error (RMSE) quantifies the typical magnitude of our prediction errors, which is invaluable for informing robust risk assessments and setting realistic expectations for portfolio management. The Symmetric Mean Absolute Percentage

Error (SMAPE) further enhances this by providing a clear, percentage-based understanding of how far off our predictions are, on average, from the actual values.

— About —

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— Purpose & Business Value —

As a passionate trader, I developed this application to streamline my decision-making process. It leverages fundamental data science concepts, including data engineering and analytics, to provide actionable insights.

The app features a user-friendly interface with a line chart, Bollinger Bands, and a Simple Moving Average (SMA) for visual analysis of price trends.

— The Model —

To enhance trading strategies and support robust decision-making, this application integrates a Prophet forecasting model, meticulously fine-tuned using advanced machine learning techniques. These methods were chosen not just for technical accuracy, but specifically to **optimize for actionable business outcomes**.

Key Model Enhancements chosen for their impact on decision-making:

- **Winsorization:** This technique was applied to improve the model's robustness against extreme price fluctuations (outliers). By mitigating the impact of unusual data points, the model generates **more stable and reliable predictions, reducing noise and leading to more confident trading decisions**. The thresholds are dynamically adjusted based on the stock's volatility to ensure relevance.
- **Adaptive Training Data:** The size of the training dataset is dynamically adjusted based on the stock's volatility and available data. This ensures the model is trained on the most relevant historical period, which is crucial for **maintaining forecast agility and relevance in fluctuating market conditions**.
- **Hyperparameter Tuning:** Through a cross-validated grid search, key model parameters (changepoint_prior_scale and seasonality_prior_scale) are systematically optimized. This process ensures the model learns the underlying patterns most effectively, leading to **highly accurate forecasts that directly translate into improved decision quality and reduced financial risk**.

— Model Configuration Options —

Training Data Duration: Users can control the amount of historical trading data used to train the forecasting model. The data consists of **trading days only** (excluding weekends and market holidays) - approximately 250 trading days per calendar year. More data generally improves accuracy by capturing longer-term patterns and trends, but increases processing time. However, more data can also lead to **model drift** - where the model becomes less sensitive to recent market changes and trends because it's heavily weighted toward older, potentially irrelevant data. The default setting uses all available data (~500 trading days $\approx$ 2 years) to provide comprehensive learning while maintaining relevance.

Monitor for model drift by checking if the forecast confidence intervals contain recent actual prices and whether the model captures recent trend changes. If the model seems disconnected from recent market behavior, try reducing the training duration.

Hyperparameter Control: While the system automatically finds optimal hyperparameters by default, users can override this automation for experimentation:

- **Changepoint Prior Scale:** Controls how flexible the model is to trend changes. Lower values create smoother, more conservative trends, while higher values allow the model to detect and adapt to more frequent trend shifts.
- **Seasonality Prior Scale:** Controls how much seasonal variation the model captures. Lower values produce smoother seasonal patterns, while higher values allow for more volatile and pronounced seasonal effects.

Note: Modifying hyperparameter values will disable automatic optimization and use your custom settings directly, which can be useful for testing specific scenarios or when you have domain expertise about the stock's behavior.

Data Preparation: Before modeling, careful data preparation steps were undertaken. While the raw data from our source (Alpha Vantage) is robust and guarantees cleanliness, I performed essential transformations such as date alignment, feature engineering (e.g., adding technical indicators), and dynamic winsorization to prepare the data optimally for the Prophet model.

By combining these refinements with a cross-validated grid search, this application provides a robust forecasting tool.

For 'SPY', optimal values are: changepoint_prior_scale: 0.001, seasonality_prior_scale: 0.010.

Cross-validation is paramount to ensuring the model's generalizability and reliability, directly translating to **trustworthiness in business insights**. By rigorously evaluating the model's

performance on multiple, unseen subsets of the data during the grid search, we can select hyperparameters that are not overfitted to a specific dataset. This robust validation process ensures that the model performs consistently on new data, providing a dependable foundation for trading decisions and strategic planning. Check out Model Iterations in the More Metrics section (above) to observe the model's improvement over its learning cycles.

— Data Source & Considerations —

The core forecasting methodology behind this application was rigorously tested and validated in a prior experiment involving 150 stocks of varying volatility, utilizing comprehensive historical data from [yfinance](#). That experiment demonstrated a Symmetric Mean Absolute Percentage Error (SMAPE) of approximately 15% across the diverse dataset, showcasing the model's general effectiveness.

For this live demonstration, market data is sourced via a free-tier API (Alpha Vantage). Due to API limitations, the available historical data for analysis and forecasting is significantly shortened compared to the original validation experiment. While this allows for a functional demonstration of the forecasting capabilities, **a diligent business intelligence analyst would note that this limited data history can impact the robustness and long-term reliability of the forecasts presented here.** The purpose remains to illustrate the application's functionality and demonstrate a comprehensive analytical workflow, rather than to provide definitive predictive accuracy based on constrained data.

— Trading Tips —

By combining the forecasting model with visual aids like a line chart, Bollinger Bands, and SMAs, I'm able to identify potential entry and exit points with greater confidence, ultimately refining my trading decisions.

By selecting a stock ticker, the app displays important background information like historical highs/lows, percentage change, volatility, and current price alongside the chart. This comprehensive tool empowers more informed trading decisions and refined trading strategies.

— Appendix —

[Click here to expand](#)

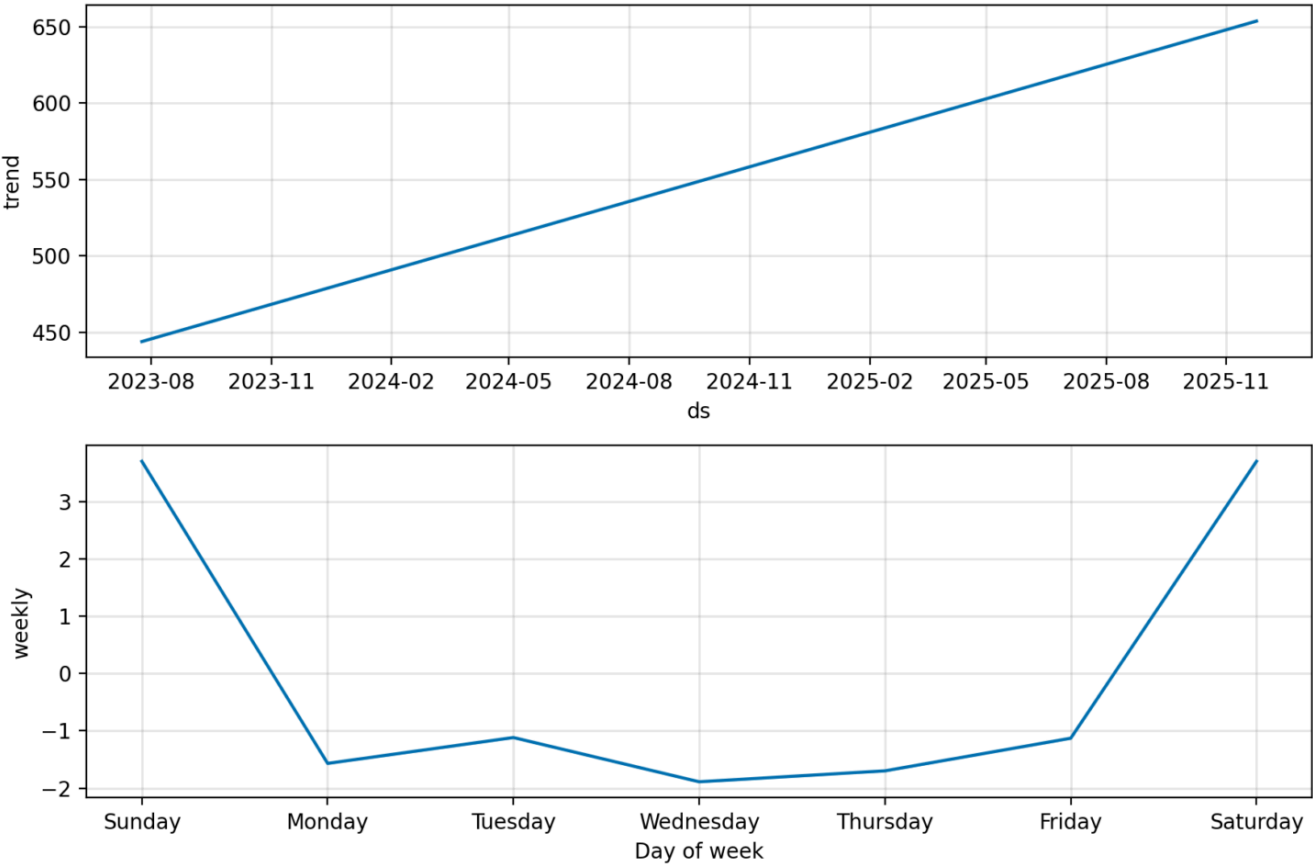


— Ticker List —

	currency	description	displaySymbol	figi	isin	mic	:
0	USD	SUMITOMO HEAVY INDUS-UNS ADR	SOHVV	BBG000JT96M8	None	OOTC	
1	USD	EMPIRE BANCSHARES INC	EBSH	BBG000GLPP48	None	OOTC	
2	USD	TEVA PHARMACEUTICAL-SP ADR	TEVA	BBG000C0CD58	None	XNYS	
3	USD	PROCEPT BIOROBOTICS CORP	PRCT	BBG009KD84H4	None	XNAS	
4	USD	SCHEID VINEYARDS INC-CLASS A	SVIN	BBG000BW6XF1	None	OOTC	
5	USD	ECOLAB INC	ECL	BBG000BHKYH4	None	XNYS	
6	USD	CATHAY PAC AIRWAYS-SPONS ADR	CPCAY	BBG000BK9JB9	None	OOTC	
7	USD	PGIM S&P 500 BUFF 20 ETF DEC	PBDE	BBG01MQT6PZ8	None	BATS	
8	USD	FIDELITY DISRUPTIVE MED ETF	FMED	BBG01GXLXWJ3	None	XNAS	
9	USD	PAGAYA TECHNOLOGIES LTD -A	PGY	BBG00VJFX5J9	None	XNAS	

— Forecast Components —

The 'Forecast Components' provide invaluable business intelligence by decomposing the overall forecast into its fundamental drivers. These components, including trend, weekly seasonality, and yearly seasonality, reveal the underlying patterns that Prophet has identified in the historical data. (Note: Yearly seasonality may be unavailable if given limited historical data.) For a business analyst, this isn't just about seeing what the forecast is, but crucially, understanding why the model predicts what it does. By visualizing these individual contributions, analysts can gain insights into long-term growth or decline, regular weekly fluctuations, and recurring annual cycles, empowering them to make more informed decisions based on a clear understanding of the forces shaping future outcomes.



— Forecast Grid —

	ds	trend	yhat_lower	yhat_upper	trend_lower	trend_upper	additive_t
0	2023-07-25 00:00:00	443.8056	410.0793	473.4554	443.8056	443.8056	-1.
1	2023-07-26 00:00:00	444.0519	411.7383	474.4505	444.0519	444.0519	-1.
2	2023-07-27 00:00:00	444.2982	411.8047	474.1765	444.2982	444.2982	-1.
3	2023-07-28 00:00:00	444.5444	412.8228	475.0835	444.5444	444.5444	-1.
4	2023-07-31 00:00:00	445.2832	411.2623	474.1381	445.2832	445.2832	-.
5	2023-08-01 00:00:00	445.5295	411.0854	475.6414	445.5295	445.5295	-1.
6	2023-08-02 00:00:00	445.7758	411.3858	475.7397	445.7758	445.7758	-1.
7	2023-08-03 00:00:00	446.022	412.4804	475.2469	446.022	446.022	-1.
8	2023-08-04 00:00:00	446.2683	413.1984	478.7727	446.2683	446.2683	-1.
9	2023-08-07 00:00:00	447.0071	413.3511	475.9553	447.0071	447.0071	-.

— Raw Data (Filtered) —

	Date	Open	High	Low	Close	Volume	Adjusted Close	daily_rei
0	2023-07-24 00:00:00	453.37	455.04	452.298	454.2	54,023,382	454.2	I
1	2023-07-25 00:00:00	453.92	456.735	453.87	455.44	55,191,240	455.44	0.
2	2023-07-26 00:00:00	454.47	456.99	453.38	455.51	71,052,885	455.51	0.
3	2023-07-27 00:00:00	459.02	459.44	451.55	452.49	92,194,375	452.49	-0.
4	2023-07-28 00:00:00	455.88	457.78	452.4918	456.92	80,011,790	456.92	0.
5	2023-07-31 00:00:00	457.41	458.16	456.045	457.79	62,040,449	457.79	0.
6	2023-08-01 00:00:00	456.27	457.25	455.49	456.48	55,502,199	456.48	-0.
7	2023-08-02 00:00:00	453.25	453.52	449.3499	450.13	93,933,369	450.13	-0.
8	2023-08-03 00:00:00	448.04	450.79	447.37	448.84	64,419,692	448.84	-0.
9	2023-08-04 00:00:00	450.72	452.895	446.27	446.81	100,128,902	446.81	-0.